1 $048 + 180$ **[M1]**

228 **[A1]**

[Total Marks: 2]

2 Identifies the relationship between alternate angles or uses $180° + 115°$ **[M1]**

295 **[A1]**

[Total Marks: 2]

3 **(a)** $180 + 58$ **[M1]**

238 **[A1]**

(b) $180 - 84 = 096$ or 096 **[B1]**

[Total Marks: 3]

4 **(a)** $068 + 180$ **[M1]**

248 **[A1]**

(b) 270 **[B1]**

[Total Marks: 3]

5 **(a)** $128 + 180$ **[M1]**

308 **[A1]**

(b) 52 **[B1]**

(c) $180 - 90 - 52$ **[M1]**

38 **[A1]**

[Total Marks: 5]

6 **(a)** $180 + 148$ OR $360 - (180 - 148)$ **[M1]**

328 **[A1]**

(b) **(i)** 270 **[B1]**

(ii) Identifies the bearing of P from B is 090° **[B1]**

$148 - 90 = 58$ **[A1]**

[Total Marks: 5]

7 **(a)** $124 + 180$ **[M1]**

304 **[A1]**

(b) $\angle PQR = 90°$ or $\angle QPR = 180° - 124° = 56°$ **[M1]**

$\angle PRQ = 180° - 90° - 56° = 34°$ **[A1]**

[Total Marks: 4]

8 (a) Bearing of P from $Q = 115 + 180 = 295°$ **[M1]**

Angle $PQR = 295° - 205° = 90°$ **[A1]**

(b) $15^2 + 8^2$ **[M1]**

$225 + 64 = 289$ **[M1]**

17 **[A1]**

[Total Marks: 5]

9 (a) Interior angle at Q on North line $= 180° - 125° = 55°$ **[M1]**

Angle $PQR = 360° - 240° - 55° = 65°$ **[A1]**

(b) $PR^2 = 45° + 60° - 2(45)(60)\cos(65°)$ **[M2]**

$PR = 57.8$ or $57.82...$ **[A1]**

[Total Marks: 5]

10 (a) 150 **[B1]**

(b) $180 + 65$ OR $360 - (180 - 65)$ **[M1]**

245 **[A1]**

(c) Identifies bearing of A from B is $245°$ (can follow through from (b)) **[M1]**

$245 - 70 = 175$ **[A1]**

(d) 25 **[B1]**

[Total Marks: 6]

11 (a) Calculates the interior angle at Q on the North side: $180° - 50° = 130°$ OR bearing of P from $Q = 180° + 50° = 230°$ **[M1]**

Calculates angle PQR: $360° - 130° - 140° = 90°$ OR $230° - 140° = 90°$ **[A1]**

(b) Uses Pythagoras' theorem: $PR^2 = 40^2 + 60^2$ **[M1]**

$72.11\cdots = 72.1$ **[A1]**

(c) Uses trigonometry to find angle QPR: $\tan(QPR) = \dfrac{60}{40}$ **[M1]**

Angle $QPR = 56.3\ldots°$ **[M1]**

Bearing $= 050° + 56.3° = 106.3°$ **[A1]**

[Total Marks: 7]

12 **(a)** Interior angle at B (South to BA) $= 35°$ OR interior angle from North line $= 180 - 35 = 145°$ **[M1]**

Angle $ABC = 360 - 145 - 125 = 90°$ OR $125 - 35 = 90°$ **[A1]**

(b) $AC^2 = 12^2 + 16^2$ **[M1]**

$AC^2 = 144 + 256 = 400$ **[M1]**

$AC = 20$ **[A1]**

(c) $\tan(\angle BAC) = \dfrac{16}{12}$ or $\sin(\angle BAC) = \dfrac{16}{20}$ **[M1]**

$\angle BAC = 53.13°$ **[A1]**

Bearing $= 035° + 53.13° = 088.1°$ **[A1]**

[Total Marks: 8]

13 **(a)** Bearing of A from B is $065° + 180° = 245°$ or Interior angle between North and AB is $180° - 65° = 115°$ **[M1]**

$245° - 155° = 90°$ or $360° - 115° - 155° = 90°$ **[A1]**

(b) Triangle ABC is right-angled and isosceles, so $\angle BCA = (180° - 90°) \div 2 = 45°$ **[M1]**

Bearing of B from C $= 155° + 180° = 335°$ **[M1]**

Bearing of A from C $= 335° - 45° = 290°$ **[A1]**

(c) $\angle DAB = 90°$ **[M1]**

Bearing of D from A $= 065° + 90° = 155°$ **[A1]**

[Total Marks: 7]

14 **(a)** Calculates angle $ABC = 240° - 170° = 70°$ (or equivalent geometric reasoning) **[B1]**

Uses the cosine rule: $AC^2 = 120° + 180^2 - 2(120)(180)\cos(70°)$ **[M1]**

$AC^2 = 14400 + 32400 - 43200(0.3420\ldots) = 32024.7\ldots$ **[M1]**

$178.9\cdots = 179$ (to 3 sf) **[A1]**

(b) Uses the sine rule to find angle BAC: $\dfrac{\sin(BAC)}{180} = \dfrac{\sin(70°)}{178.95}$ **[M1]**

$\sin(BAC) = 0.9452\cdots \implies$ Angle $BAC = 70.9°$ **[M1]**

Bearing $= 060° + 70.9° = 130.9°$ **[A1]**

[Total Marks: 7]

15 **(a)** $\angle QPR = 118° - 42° = 76°$ **[M1]**

$QR^2 = 24^2 + 35^2 - 2(24)(35)\cos(76°)$ **[M1]**

$QR = 37.3$ or $37.34\ldots$ **[A1]**

(b) $\dfrac{\sin(\angle PQR)}{35} = \dfrac{\sin(76°)}{37.34}$ OR $\cos(\angle PQR) = \dfrac{24^2 + 37.34° - 35^2}{2(24)(37.34)}$ **[M1]**

$\sin(\angle PQR) = 0.909\ldots$ OR $\cos(\angle PQR) = 0.415\ldots$ **[M1]**

$\angle PQR = 65.4°$ or $65.42\ldots°$ **[A1]**

Bearing $= 180 + 42 - 65.4 = 156.6°$ (or $157°$) **[A1]**

[Total Marks: 7]

16 (a) Angle $ABC = 360° - 145° - (180° - 42°) = 77°$ **[M1]**

$\dfrac{\sin(BAC)}{BC} = \dfrac{\sin(77°)}{18}$ **[M1]**

$BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(BAC)$ (Implicitly requires finding BC or BAC first. Correct approach: $\dfrac{\sin(BCA)}{12} = \dfrac{\sin(77°)}{18}$) **[M1]**

$\sin(BCA) = \dfrac{12\sin(77°)}{18} = 0.6495...$ **[A1]**

Angle $BCA = 40.5°$ **[A1]**

Angle $BAC = 180° - 77° - 40.5° = 62.5°$ **[M1]**

Bearing $= 042° + 62.5° = 104.5°$ **[A1]**

(b) Boat sails on perpendicular bisector of BC. Angle $BAC = 62.5°$. The bisector of angle BAC is not necessarily the perpendicular bisector of BC.

Line is perpendicular bisector of BC.

Let midpoint of BC be M. In $\triangle ABM$, AM is the path.

Use cosine rule to find BC: $BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(62.5°)$

BC = 16.39...

$\sin(ABM)/AM = \sin(BAM)/8.19...$ (This approach is overly complex).

Simpler: Path is perpendicular bisector. Angle of BC with North is 145°.

Perpendicular to BC has bearing $145° - 90° = 055°$.

Recognise path is perpendicular to BC **[M1]**

Bearing $= 145 - 90 = 055$ (or $055°$) **[A1]**

[Total Marks: 7]

17 (a) Bearing of P from $A = 050 + 180 = 230°$ **[M1]**

Angle $PAB = 230° - 160° = 70°$ **[M1]**

$PB^2 = 40^2 + 60^2 - 2(40)(60)\cos(70°)$ **[M1]**

59.6 or $59.65\ldots$ **[A1]**

(b) Uses sine rule or cosine rule to find angle APB **[M1]**

$\dfrac{\sin(APB)}{60} = \dfrac{\sin(70°)}{59.65}$ or $\cos(APB) = \dfrac{40^2 + 59.65^2 - 60^2}{2(40)(59.65)}$ **[M1]**

Angle $APB = 70.9°$ $(70.93\ldots°)$ **[A1]**

Bearing $= 050 + 70.9 = 120.9°$ **[A1]**

[Total Marks: 8]

18 **(a)** Identifies angle $PQR = 230° - 120° = 110°$ (or uses parallel lines to find $180 - 50 = 130$ and $360 - 130 - 120 = 110$) **[B1]**

Uses Cosine rule: $PR^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ [M2] (M1 for correct substitution into formula)

$PR = 86.4$ or $86.44\ldots$ **[A1]**

(b) Uses Sine rule: $\dfrac{\sin(QPR)}{60} = \dfrac{\sin(110°)}{86.44}$ **[M1]**

Angle $QPR = 40.7°$ $(40.73\ldots)$ **[A1]**

Bearing $= 050° + 40.7° = 090.7°$ **[A1]**

(c) $090.7 + 180$ **[M1]**

270.7 **[A1]**

[Total Marks: 9]

19 **(a)** 090
090 **[B1]**

(b) 054
+
180
054
+180

[M1]

234
234 **[A1]**

(c) Angle
=
90
54
=
36
BAC=90
54
=36
and angle
=
180
36
2
=
72
ABC=
2
180
36

=72

[M1]

Bearing
=
234
72
=
162
=234
72
=162 **[A1]**

[Total Marks: 5]

20 (a) 065
065 (accept
063
063 to
067
067) **[B1]**

(b) Measured length
=
6
PQ=6 cm (accept
5.8
5.8 cm to
6.2
6.2 cm) **[B1]**

30
30 (accept
29
29 to
31
31) **[A1]**

(c) Line drawn from
Q on bearing
140
140
(accept
138
138
to
142
142
) **[B1]**

Point
R marked at distance
5
5 cm from
Q (accept

4.8
4.8 cm to
5.2
5.2 cm) and labelled **[B1]**

(d) 065
+
180
065
+180

[M1]

245
245 **[A1]**

[Total Marks: 7]

21 **(a)** 270
270 **[B1]**

(b) 018.6
+
180
018.6
+180

[M1]

198.6
198.6 **[A1]**

(c) **(i)** 198.6
48.4
198.6
48.4

[M1]

150.2
150.2 **[A1]**

(ii) Angle
=
90
18.6
=
71.4
ADC=90
18.6
=71.4

[B1]

Angle
=
322.8
270
=
52.8

BCD=322.8
270
=52.8

[B1]

Angle
=
360
(
71.4
+
150.2
+
52.8
)
=
85.6
ABC=360
(71.4
+150.2
+52.8
)=85.6

[A1]

(d) Bearing of
A from
=
048.4
+
180
=
228.4
B=048.4
+180
=228.4

[M1]

Bearing of
C from
=
228.4
85.6
=
142.8
B=228.4
85.6
=142.8

[A1]

[Total Marks: 10]